BUILD BLUEPRINT / VERSION 2

BTU to Tons Developer Design & SEO Spec

A clear interface, a reliable calculation engine, and a focused search architecture.

CONVERT FIRST. EXPLAIN THE RESULT. HELP WITH THE NEXT STEP.

One shared engine powers the main calculator, numeric shortcuts, batch conversion and comparison. Room sizing, nameplate guidance and power estimates are clearly separated so users understand what each result means.

01 / DESIGN

Desktop and mobile wireframes, section order, controls and ready-to-use copy.

02 / ENGINE

Formulas, unit conventions, validation and expected test outputs.

03 / SEARCH

20 numeric shortcuts, 12 supporting routes and page-level SEO copy.

04 / EVIDENCE

Seven named competitors reviewed; revised claims and source-linked decisions.

Prepared for: Hasnain Nawaz and the development team
Inputs: supplied landing-page blueprint + 2,487-row keyword CSV
Review date: 10 October 2026

This document supersedes the supplied blueprint. Domain and visual branding are not specified; paths are proposed relative URLs. Colors shown are an implementation concept, not an existing brand claim.

01 / RECONCILIATION

Read this before building

Old direction	Reconciled implementation
“Nobody combines sizing and conversion.”	NumberVibe already does. Compete on clearer results, batch work, comparison and explicit assumptions. [C3]
8 competitors / 3,102 keyword rows	Seven competitors are identifiable by name; the eighth is unnamed. Only the 2,487-row CSV is supplied. The 615-row FAQ dataset is unavailable.
Every BTU keyword belongs on one page	Separate conversion, sizing, definitions, electrical estimates and unrelated meanings. Do not create a URL for each spelling variant.
20–25 BTU per sq ft + climate multipliers	Replace with a sourced room-AC lookup. Do not label ad hoc factors “Manual J” or “ASHRAE compliant.” [S2–S4]
Any model containing 036 = 3 tons	Accept explicit rated capacity first. Decode only documented model families; ambiguous matches return no answer. [S5]
Link to watts-to-amps to size a breaker	Cooling kW is not electrical input. Never derive an HVAC breaker from tonnage; use nameplate and installation documentation. [S5–S7]
FAQ schema creates search enhancement	Google says FAQ rich results ended May 7, 2026. Keep useful visible FAQs; remove the rich-result promise. [S8]

Build sequence

P0: core converter, selected shortcut pages, comparison, batch conversion and explanatory content. P1: room sizing and power/energy utilities. P2: verified model-family lookup and larger editorial cluster. No automatic heating-size recommendation at launch.

Navigation: product and screens pp. 3–14; engine and copy pp. 15–18; shortcut templates and SEO pp. 19–26; technical implementation and rollout pp. 27–28; source register pp. 29–30.

02 / SEARCH SCOPE

Keyword coverage without intent mixing

The supplied CSV has **2,487 unique keywords**, including **569 zero-volume rows**. Its summed Volume column is **350,040**; this is overlapping keyword demand, not unique visitors or a traffic forecast. Export geography and collection date are unspecified.

Routing bucket	Rows	Decision / destination
Definitions / broad BTU intent	275	Main summary + future /what-is-btu/ guide.
Other power units / ambiguity	122	Unit routes + ambiguity checks.
General HVAC / editorial review	312	Review individually before assigning a publishable URL.
Room sizing / load intent	455	/room-ac-size-calculator/
Core / reverse conversion	593	Main and reverse routes.
Other meanings / manual review	97	Exclude from refrigeration converter; separate qualified tools only.
Heating / separate scope	91	Heating guide; no generic cooling-to-heating toggle.
Nameplate / tonnage guide	21	/ac-tonnage-by-model-number/
Numeric capacity shortcuts	502	20 initial numeric pages; other values use the core tool.
Thermal energy / ton-hours	14	/btu-to-ton-hours/
Specialist refrigeration / defer	5	Explain distinction; defer specialist calculators.

Method: ordered, mutually exclusive text-pattern routing; each source row receives one provisional bucket. Counts are reproducible classifications, not manually validated intent labels. Ambiguous terms remain in review instead of being forced onto a page.

Verified examples: “btu to tons” 1,900; “24000 btu to ton” 210; “36000 btu to tons” 210; “btu per square foot” 2,400; “how to determine ac tonnage from model number” 260. Use variants naturally; avoid repeating all 2,487 strings in copy.

03 / PAGE-LEVEL REVIEW

Competitors: what is actually present

Review scope: retrieved public page text and exposed controls on 10 October 2026. This is not a browser interaction test, a speed audit, a code/schema audit or a location-specific ranking study. “Not observed” does not establish site-wide absence.

Page	Observed strengths	Gap / response
Omni [C1]	BTU, tons, watts; sharing; definitions; author/reviewer profiles; FAQ.	No inline room tool or model decoder observed. Separate thermal output from electrical input more explicitly.
RapidTables [C2]	BTU/hr input; result; inverse link; formula and compact table.	No inline batch, comparison or room-sizing controls observed. Keep its low-friction conversion flow.
NumberVibe [C3]	Conversion + room modes; presets; multiple units; precision; cost inputs; size table.	The old uniqueness claim fails. No documented model-family decoder or batch import observed.
Calculator Academy [C4]	BTU/hr, W, kBTU/hr, TR, kW; unit clarity; large capacity chart; examples.	Previous 3-use gate not confirmed in retrieved page. Do not market against an unverified restriction.
CtrlCalculator [C5]	Simple input; numeric table; worked examples; time-basis reminder.	No batch or interactive room tool observed. Its area examples are not a load-calculation method.
FreeConvert [C6]	Automatic conversion prompt and unit selector.	Large general-purpose navigation. Retrieval exposes limited calculator detail; dynamic capabilities remain uncertain.
CreativeBooster [C7]	Reverse conversion; long numeric chart; linked numeric pages.	Prose confuses refrigeration tonnage with mass. Correct entity definitions and meaningful shortcut pages are an opportunity.

Avoid copying competitor sizing factors. Observed features establish a baseline; authoritative sources should establish the mathematics. All competitor URLs appear in the source register.

04 / OPPORTUNITY MATRIX

Differentiation worth implementing

Feature	Competitive status	Build decision
Core + reverse + kW	Established across multiple pages.	P0 baseline. Fast, complete and unit-correct.
Room sizing + costs	Already present at NumberVibe.	P1 parity. Win through transparent method and limitations.
Numeric shortcut pages	Already present at CreativeBooster.	P0 selective. Distinct answers, meaningful comparisons and editable presets.
Batch paste + CSV export	Not observed in reviewed page content.	P0: up to 200 values, line-level errors, full-precision calculation.
Compare A vs B	Dedicated side-by-side workflow not observed.	P0: absolute and relative capacity difference; no equipment recommendation.
Explicit time basis	Some pages explain /hr; minute/second workflows not prominent.	P1: /min, /s, energy + duration and ton-hours with distinct labels.
Verified model lookup	Documented family-specific lookup not observed.	P2: source URL, family scope, exact match and uncertain-result state.
Nominal vs rated result	Not prominent as a decision workflow.	P0 explanation; P2 display both when manufacturer data supports them.
Electrical nameplate checklist	Not observed as an integrated workflow.	P1: MCA/MOCP education and manufacturer-document handoff; no automatic breaker size.

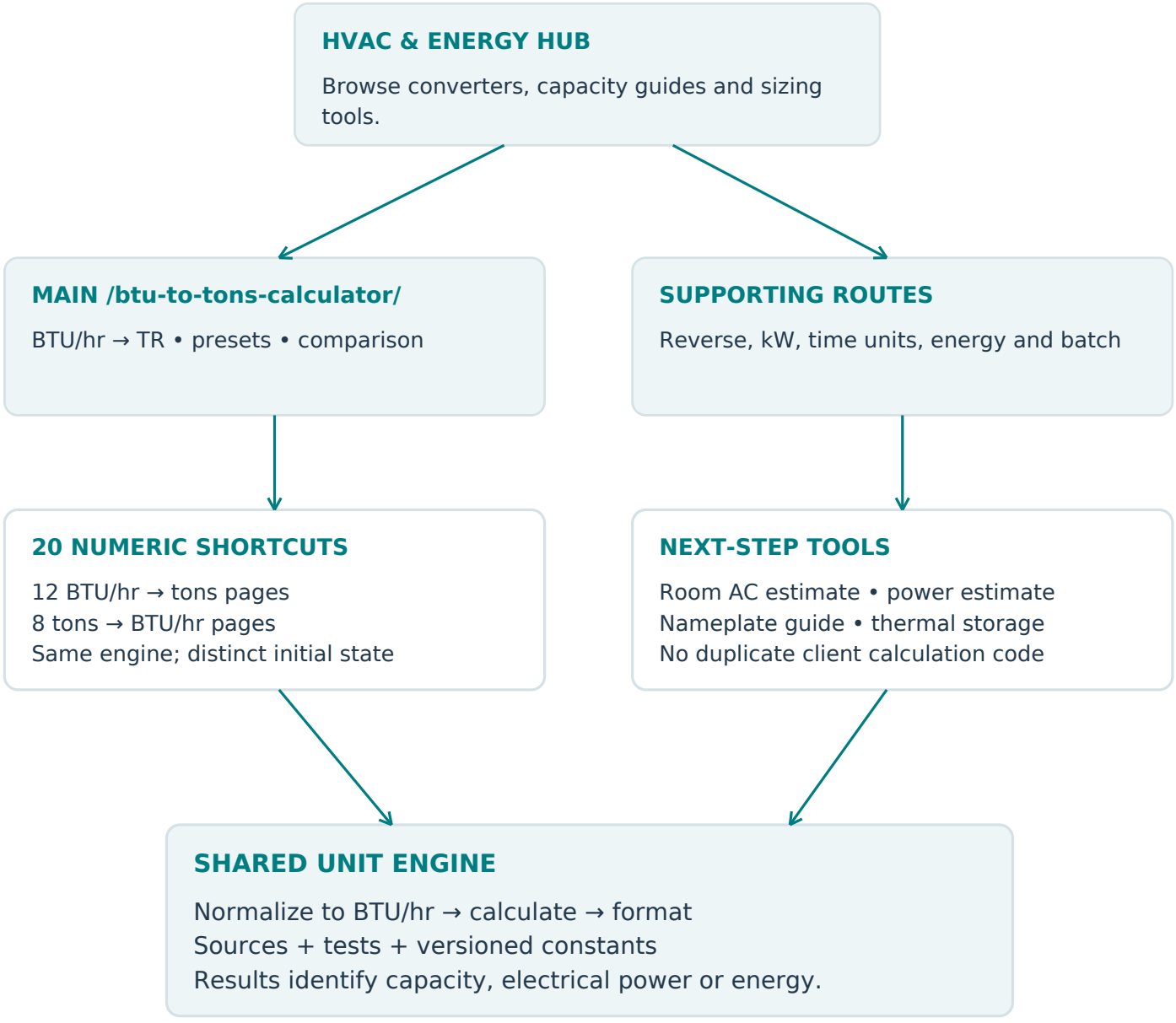
Positioning copy

Know the capacity. Understand the difference.
Convert one rating, compare two systems or work through a full list. See the formula and units behind every result.

These are scoped product opportunities, not a claim that no calculator anywhere offers them. Recheck the benchmark before using public “first,” “only” or “best” claims.

05 / INFORMATION ARCHITECTURE

One engine, several useful entry points



Internal links: main chart → relevant numeric page; numeric page → main + reverse equivalent + two neighboring capacities; result → power estimate; room guide → main converter; electrical guide → watttoamp.com using “convert known electrical watts to amps.”

URL rule: paths in this document assume a multi-tool domain. If a dedicated domain uses its homepage as the main converter, use “/” as canonical and redirect the duplicate calculator path. Never index both identical routes.

06 / WIREFRAME A

Full landing page: section order

Numbers are component IDs used in the copy and implementation specifications.

01 / Header Logo • Convert • Charts • Guides	Compact navigation; no login requirement.
02 / Hero + converter H1 • input • units • result • copy / share	First usable tool visible immediately.
03 / Quick capacities 6k / 9k / 12k / 18k / 24k / 36k	Buttons prefill; “details” links open indexable pages.
04 / Compare + batch Compare two ratings | Convert a list	Collapsed utilities; keep primary form simple.
05 / Reference chart BTU/hr | tons | cooling kW	Searchable HTML table; shortcuts linked.
06 / Meaning + method What is BTU? • Why tons? • example	Brief answers before deeper guidance.
07 / Room sizing Room AC estimator preview + open tool	Independent form; never alters converter output.
08 / Nameplate guide Read capacity | supported model lookup	Explicit unknown and ambiguous states.
09 / Related estimates Electrical input • ton-hours • unit shortcuts	Load tool only after user chooses it.
10 / Trust + FAQ Method • reviewer • limitations • questions	Visible, server-rendered copy; no FAQ rich-result claim.
11 / Footer Related tools • contact • privacy • terms	Only link to available pages.

Desktop max-width: 1120px. Main body: 24px gap, 16–24px card padding, 16px text minimum. Keep the page focused; supporting full tools can live on dedicated routes.

07 / WIREFRAME B

Desktop converter: exact UI

BTU to Tons Calculator

Convert cooling capacity. 12,000 BTU/hr = 1 refrigeration ton.

[BTU/hr to Tons] [Tons to BTU/hr] [More units]

Cooling capacity

[24,000]
[BTU/hr v]
Example value; editable

YOUR CONVERSION

2 refrigeration tons
24,000 BTU/hr • 7.0337 cooling kW
Exact unit ratio; displayed kW rounded

QUICK VALUES

[6,000] [9,000] [12,000] [18,000] [24,000] [36,000]

SHOW THE MATH

24,000 ÷ 12,000 = 2 tons
[Copy result] [Share calculation] [Reset] [Precision: 4]

Cooling kW is not electricity use.

To estimate electrical input, enter an efficiency rating or measured input watts.

Control	Required behavior
Input / units	Live update after valid edit; Enter also calculates. Unit changes preserve the quantity, not the digits.
Direction tabs	Switch view and units, preserving the current physical capacity. Link to reverse page via a separate crawlable link.
Copy / share	Copy value + units + equation. Share a URL fragment containing only approved numeric state.
Reset / precision	Reset to blank form. Choose 2, 4 or 6 decimal places; never round intermediate math.

08 / WIREFRAME C

Mobile: one task at a time

BTU to Tons Calculator

Convert cooling capacity.

[BTU → Tons] [Reverse]

Cooling capacity

[24,000]
Unit: BTU/hr

2 refrigeration tons

24,000 BTU/hr
7.0337 cooling kW

[Copy] [Share] [Reset]

24,000 ÷ 12,000 = 2

Show formula and rounding details

Compare / Batch / Chart

Read order

H1 → direction → input → result → actions → presets. Avoid horizontal scrolling.

Touch controls

44px minimum target; visible labels; numeric keyboard with decimal entry.

Accessible results

Polite status announcement after validation. Preserve focus; do not announce each keystroke.

Reference chart

Three columns fit mobile. Row details open underneath; headers remain meaningful.

Errors

Inline explanation below the field. Hide stale result after an invalid edit.

Small screens

One-column layout below 768px; no fixed-height cards in production. Support 200% zoom.

Wireframes show layout intent, not pixel-perfect final branding. Implement semantic HTML controls and test at 360px, 390px, tablet and desktop widths.

09 / WIREFRAME D

Batch and comparison workflows

COMPARE CAPACITY

A: [18,000] [BTU/hr]
B: [24,000] [BTU/hr]

B has 6,000 BTU/hr more capacity.
Difference: 0.5 tons / 1.7584 kW
B is 33.33% higher than A.

[Use A] [Use B] [Copy comparison]

BATCH CONVERT

Input unit: [BTU/hr]
[12000
18000
24000]
One number per line; up to 200.

[Convert list] [Download CSV]

Behavior	Exact requirement
Comparison math	Normalize both inputs. Difference = B - A; percent = 100 × (B - A) / A. If A = 0, show “Percentage unavailable: baseline is zero.”
Batch parser	One value per line. Permit valid thousands separators within a value; no comma-separated lists. Trim blank lines. Reject ambiguous decimal commas.
Mixed units	One selected input unit for the whole batch at launch. Tell users to split lists with different units.
Errors	Preserve row order. Show row number and reason for invalid values; calculate valid rows. Never silently delete rows.
Export	Columns: row, input_value, input_unit, BTU_per_hour, refrigeration_tons, cooling_kW, status. Export numbers at up to 12 significant digits.
Privacy	Process locally. No list or model string in analytics; explicit action required to copy or download.
Comparison copy	“Capacity difference only. Efficiency, room suitability and actual performance are separate.”

No new standalone comparison URL at launch: /btu-to-tons-calculator/#compare. Batch has a separate indexable page because its workflow and instructions are materially different.

10 / WIREFRAME E

Room AC estimate: a bounded method

ROOM AIR CONDITIONER SIZE

Area [300] [ft² / m²]
Sun [Normal / Shaded / Sunny]
People [2] Kitchen [No]
Ceiling approximately 8 ft? [Yes]
Use [Single room / Whole home]

[Estimate room capacity]

PRELIMINARY ROOM ESTIMATE

8,000 BTU/hr
0.6667 refrigeration tons

Method: ENERGY STAR room-AC table.
For room/window AC guidance only.
[See assumptions] [Convert result]

Scope: use the ENERGY STAR area table for room air conditioners, approximately 8-ft ceilings. Do not apply it as a central-system Manual J calculation. Whole-home, heating, unusual ceiling height or out-of-range area returns a professional-load-calculation guide. [S2–S4]

Area ft² → base BTU/hr	Area ft² → base BTU/hr
100–<150 → 5,000	550–<700 → 14,000
150–<250 → 6,000	700–<1,000 → 18,000
250–<300 → 7,000	1,000–<1,200 → 21,000
300–<350 → 8,000	1,200–<1,400 → 23,000
350–<400 → 9,000	1,400–<1,500 → 24,000
400–<450 → 10,000	1,500–<2,000 → 30,000
450–<550 → 12,000	2,000–2,500 → 34,000

Implementation: ft² = m² × 10.7639104167. Apply sun multiplier 0.9 / 1 / 1.1 to the base, then add 600 × max(people - 2, 0), plus 4,000 for a kitchen. This operation order is our explicit implementation convention. Occupants must be a nonnegative integer. [S2]

Do not round this estimate to a recommended central AC tonnage. Show the table result and assumptions. Source attribution is not ENERGY STAR certification or endorsement.

11 / WIREFRAME F

Nameplate guide: confidence before output

FIND YOUR AC CAPACITY

[I have a BTU/hr rating] [I have a model number]

EXPLICIT CAPACITY / P0

Rated cooling capacity [35,000]
[BTU/hr / cooling kW]

Result: 2.9167 tons
Label: Converted rated capacity

DOCUMENTED LOOKUP / P2

Brand [Select] Family [Select]
Model [Exact model identifier]

[Check supported documentation]
Result includes source + nominal label

State	Copy / behavior
Known rating	“Enter the cooling-capacity rating, not power input or refrigerant charge.” Convert it using the normal engine.
Supported match	“Nominal capacity: 3 tons. Confirm the rated capacity in the manufacturer performance table.” Show exact source page and model-family rule.
Unknown	“We cannot verify this model. Enter the rated BTU/hr value or check the manufacturer documentation.”
Ambiguous	“More than one capacity code matched. No tonnage has been assigned.” Do not take the first 3-digit match.
Data structure	brand, family, region, model-pattern, capacity-code-position, nominal-BTU/hr, source-URL, source-page, verification-date, reviewer.
Privacy / scope	No serial number required. No OCR upload in launch scope. Never apply one brand’s model pattern to another brand.

Manufacturer example: Carrier’s 38TKB documentation distinguishes nominal size codes, combination ratings and electrical data. This supports a constrained family-specific guide, not a universal “find 036” parser. [S5]

Release the reading guide before the automatic lookup. Keep lookup disabled until a maintained, reviewed model-family dataset exists.

12 / WIREFRAME G

Power and thermal energy: separate tools

ELECTRICAL INPUT ESTIMATE

Cooling capacity [24,000 BTU/hr]
Method [EER / COP / Known watts]
EER [12] Hours/day [8]
Rate [0.15] Currency [USD]

2.000 kW electrical input
16 kWh/day • 2.40 USD/day

THERMAL ENERGY / TON-HOURS

Energy [24,000] [BTU]
Optional duration [2] [hours]

2 ton-hours of thermal energy
Average capacity over 2h: 1 ton

This is not electric meter consumption.

Tool	Formula and boundary
EER method	Electrical W = cooling BTU/hr ÷ EER. EER must be in BTU/Wh and positive; COP is a different unit convention. [S7]
COP method	Electrical kW = cooling kW ÷ COP. Require a positive cooling COP measured at compatible conditions.
Known watts	Use input watts directly for energy and cost. Do not infer cooling capacity without an efficiency rating.
Energy and cost	Electrical kWh = electrical kW × equivalent full-load hours. Cost = kWh × rate. Daily hours 0–24; rate ≥ 0. No invented seasonal load profile.
Ton-hours	Thermal ton-hours = BTU ÷ 12,000. Average TR = ton-hours ÷ duration_hours; duration must be greater than zero.
Seasonal limits	Do not substitute SEER/SEER2 for instantaneous EER. Inverter part-load behavior and weather change real consumption.

Ready-to-use helper: “Use an EER in BTU/Wh or a cooling COP in W/W. Do not enter a SEER rating here. Estimates assume constant operation at the entered efficiency.”

13 / REPLACEMENT FOR A BREAKER GUESS

The right standards-based extra

For this calculator, the useful equivalent of an NEC reference is a **capacity-to-selection checklist**: convert the rating, establish the building load, verify equipment performance, then check installation requirements. Tonnage alone cannot specify a breaker.

Reference	Feature to implement	Limit
NIST [S1]	Visible unit definition and conversion factor in every result.	The ratio converts capacity; it does not select equipment.
ENERGY STAR [S2]	Room-only sizing method with its table and stated assumptions.	No claim of certification or whole-home design.
ACCA Manual J [S3]	“Have a load report?” Enter total cooling BTU/hr and optionally sensible/latent loads.	Convert supplied loads; do not claim a simplified form performs Manual J.
ACCA Manual S [S4]	“Check rated performance at design conditions.” Link to selection guidance.	No universal capacity oversize percentage or automatic next-size-up rule.
Manufacturer [S5]	Display rated/nominal distinction; ask for the exact installation manual.	Model matches, voltage and equipment combinations vary.
NFPA 70 [S6]	Explain that HVAC electrical requirements need the adopted code and nameplate.	No breaker/wire calculator from tons. Article 440 review belongs in a separate qualified electrical workflow.

Optional load-report panel / P1

Fields: total cooling load BTU/hr; optional sensible and latent loads. Show converted TR and cooling kW. If both components are present, verify their sum equals the total within max(1 BTU/hr, 0.1%). Mismatch copy: “Check the values in your load report.” Do not label equipment as correctly sized.

Electrical checklist copy

> “Cooling capacity does not determine breaker size. Check the equipment nameplate and installation instructions for minimum circuit ampacity (MCA) and maximum overcurrent protection (MOCP). Have a qualified installer apply the locally adopted electrical code.”

The NFPA landing page exposed no code text in this review. Clause-level compliance is not asserted; draft code-development documents are not treated as adopted requirements.

14 / FORMULAS & UNITS

Calculation engine contract

Input unit	Normalize to BTU/hr	Example → TR
BTU/hr	x	24,000 → 2
Refrigeration ton / TR	x × 12,000	2 → 2
Cooling kW	x × 1,000 / 0.2930710701722222	3.5168528420667 → 1
Cooling watts	x / 0.2930710701722222	3,516.8528420667 → 1
kBTU/hr or MBH	x × 1,000	24 → 2
MMBTU/hr	x × 1,000,000	0.024 → 2
BTU/min	x × 60	200 → 1
BTU/s	x × 3,600	3.3333333333333 → 1

Output: TR = normalized BTU/hr ÷ 12,000; cooling kW = normalized BTU/hr × 0.2930710701722222 ÷ 1,000. Use the International Table BTU convention. The watt factor is based on 1 BTU_IT = 1,055.05585262 joules; NIST's published TR value is rounded. [S1]

Ambiguity rules

A bare "BTU" is energy. The primary interface always labels its capacity field "BTU/hr," even where the search phrase says BTU. In a total-energy tool, require duration for a capacity result. A bare "MMBTU" must not silently become MMBTU/hr. "Tons" always means refrigeration tons on capacity screens, never metric mass or refrigerant charge.

Numerical rules

Accept finite decimal values from 0 to 10^{12} in the selected capacity unit; this is a product limit, not a physical boundary. Reject negative, Infinity, NaN, scientific notation and mixed strings at launch. Normalize first; maintain full numeric precision; round only display/export. Show "less than 0.0001 TR" rather than zero for tiny positive values at four decimals.

Do not expose horsepower, cooling-tower tonnage, fuel mass or CO_2 conversions as equivalent HVAC units. These require additional definitions or inputs. kcal/hr is deferred to avoid silently mixing calorie conventions.

15 / ACCEPTANCE CRITERIA

Validation and release tests

Test	Expected result
24,000 BTU/hr	2 TR; 7.0337056841 cooling kW before display rounding.
1.5 TR	18,000 BTU/hr; 5.2752792631 cooling kW.
200 BTU/min / 3.3333333333333 BTU/s	Both approximately 1 TR.
24,000 BTU total over 2 hours	2 ton-hours; average 1 TR.
24,000 BTU/hr; EER 12; 8h; rate 0.15	2 kW electrical; 16 kWh; 2.40 cost units.
300 ft², normal sun, 2 people, no kitchen	8,000 BTU/hr preliminary room estimate.
300 ft², sunny, 4 people, kitchen	8,000 × 1.1 + 1,200 + 4,000 = 14,000 BTU/hr.
149.99 / 150 / 2,500 / 2,500.01 ft²	5,000 / 6,000 / 34,000 / unsupported area.
A=18,000, B=24,000 BTU/hr	B-A = 6,000; 0.5 TR; +33.33% versus A.
Blank / -1 / 12abc / 1,5	No result; explain required number, nonnegative value or format.
Zero / zero EER / zero duration	0 capacity valid; EER and duration zero invalid.
Unknown model or multiple matches	No automatic tonnage; request a rated capacity or source.

UI and technical pass conditions

Copy includes units; reset clears state; keyboard tabs follow visual order; comparison handles zero baseline; 200-line batch works and 201 lines yields a clear limit message. Unit swaps preserve physical quantity. Fragment links reproduce result without creating indexable variants.

Server-render H1, formula, default examples, chart and copy. Test client/server consistency and shortcut initial values. Numeric tests: absolute tolerance ≤ 10^{-8} TR for the ordinary cases above. Mobile tests: no clipping at 360px or 200% zoom.

16 / COMPONENT COPY

Landing page copy: ready to use

ID / heading	Final copy
02 / H1	**BTU to Tons Calculator**
02 / Subheading	Convert BTU per hour to refrigeration tons instantly. Compare cooling capacities, see the formula and check common AC ratings.
02 / Result note	One refrigeration ton equals 12,000 BTU/hr. Cooling kW describes heat-transfer capacity, not electricity consumption.
03 / H2	**Common BTU-to-ton conversions** Choose a capacity to fill the calculator, or open its worked example.
04 / H2	**Compare two ratings or convert a list** Check the difference between two capacities, or paste multiple values for a downloadable conversion table.
05 / H2	**BTU to tons conversion chart** These are equivalent capacities. A matching tonnage does not establish whether a unit is right for your room.
06 / H2	**What do BTU/hr and refrigeration tons mean?** BTU is a unit of thermal energy. BTU/hr measures the rate of heat transfer. A refrigeration ton is another capacity unit, equal to 12,000 BTU/hr; it is not the unit’s weight.
06 / Formula	**How to convert BTU/hr to tons** Divide BTU/hr by 12,000. For example, 24,000 ÷ 12,000 = 2 tons. To reverse the conversion, multiply refrigeration tons by 12,000.
07 / H2	**Need a room air conditioner size estimate?** Use a room-specific sizing method with area, sunlight and occupancy. Whole-home systems require a separate load calculation.
08 / H2	**Find the capacity on your AC nameplate** Use the cooling-capacity rating when available. Model-number codes vary by manufacturer and product family.
09 / H2	**Go beyond the conversion** Estimate electrical input with an efficiency rating, convert thermal energy to ton-hours or change the input time unit.

10 / Trust block: “Method: standard refrigeration-ton conversion using BTU_IT units. Calculations are shown for reference. Room and power estimates use separate assumptions. Method updated: [actual release date].” Publish a reviewer name only after that person has reviewed the tool.

17 / SUPPORTING CONTENT

FAQ copy and reference-chart content

Question	Answer
How many BTU are in one ton?	One refrigeration ton equals 12,000 BTU per hour. Confirm that the rating is a rate, not total energy.
Is 24,000 BTU a 2-ton AC?	A 24,000 BTU/hr cooling rating equals 2 tons. Actual rated performance can differ from a nominal model size.
Are tons the weight of the AC?	No. Refrigeration tons measure heat-transfer capacity, not equipment weight or refrigerant mass.
Does 1 ton use 3.517 kW of electricity?	No. About 3.517 kW is its cooling capacity. Electrical input depends on efficiency and operating conditions.
Can I convert total BTU to tons?	Provide a duration: divide BTU by hours, then divide by 12,000. Without time, total BTU converts to ton-hours.
How many square feet can 2 tons cool?	There is no universal area-to-tonnage ratio. Room conditions, building design and climate affect the required capacity.
Can I size a whole-home AC here?	The room estimate is not a whole-home load calculation. Use a qualified load assessment and equipment performance data.
Can the model number identify tonnage?	Sometimes. Use a documented rule for the exact manufacturer and product family; unsupported models return no estimate.
Does this work for heat pumps?	It converts a stated capacity rating. Heating and cooling capacities can differ and change with outdoor conditions.
Can tonnage tell me the breaker size?	No. Use the nameplate and installation requirements; capacity is not electrical input or circuit protection.

Chart rows: show BTU/hr, TR and cooling kW

Use 6,000; 8,000; 9,000; 10,000; 12,000; 18,000; 24,000; 30,000; 36,000; 42,000; 48,000; 60,000. Generate results from the shared engine. Label these “common capacity examples,” not a promise that all values are sold in every equipment category.

Replace the old tons-to-square-feet table with the sourced room-AC method. Keep FAQs in normal HTML; FAQ structured data is not a Google rich-result opportunity. [S8]

18 / REUSABLE PAGE CONTRACT

Shortcut page template and SEO rules

H1: 24,000 BTU to Tons

Answer first: 24,000 BTU/hr equals 2 refrigeration tons.

PREFILLED CALCULATOR

Value: 24,000 • Unit: BTU/hr
Editable live result
No automatic navigation after edits

WORKED EXAMPLE

24,000 ÷ 12,000 = 2 TR
Cooling capacity: 7.0337 kW
Separate link to electrical estimate

CONTEXT AND LINKS

H2: How the conversion works
H2: Compare 24,000 with 18,000 and 30,000 BTU/hr
H2: Does 2 tons mean it fits my space?
Links: main tool • reverse equivalent • neighboring capacities

Each numeric page must include: an immediate answer, an editable preset, a worked equation, a three-row comparison table and a concise capacity-vs-sizing explanation. BTU shortcut H1s match query language; first sentence explicitly adds “per hour.”

Shared metadata rules: one descriptive H1, self-referencing canonical, unique title and description, breadcrumb to the main converter. Slugs use digits without commas; decimal ton values use hyphens. Singular/plural spelling variants redirect to the same destination.

CTR approach: put the searched number and answer early; use an honest bracket such as [Formula] or [Chart]. A hook is a test hypothesis, not a guarantee. Avoid fake accuracy, “best,” unearned review stars or dates that change without a substantive update.

Indexation gate: publish only the curated pages with useful content. Other numeric values remain converter states. Do not mass-publish all integer permutations or keyword spellings. Reconsider a page if its only difference is one substituted number.

Pages 20–23 specify all 20 numeric launch candidates. Pages 24–26 specify all 12 supporting route candidates, including later phases. Main-page metadata appears on page 27. Deferred concepts on page 28 have no launch URL.

19 / READY-TO-IMPLEMENT SEO

Numeric shortcuts 1–5

/6000-btu-to-tons/

Title: 6,000 BTU to Tons = 0.5 [Formula + Chart]
Description: Convert 6,000 BTU/hr to 0.5 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 6,000 BTU to Tons
Answer copy: 6,000 BTU/hr equals 0.5 refrigeration tons. Divide 6,000 by 12,000 to convert the capacity.
Keyword family: 6000 btu to tons; 6000 btu to ton; how many tons is 6000 btu

/8000-btu-to-tons/

Title: 8,000 BTU to Tons = 0.6667 [Formula + Chart]
Description: Convert 8,000 BTU/hr to 0.6667 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 8,000 BTU to Tons
Answer copy: 8,000 BTU/hr equals 0.6667 refrigeration tons (rounded). Divide 8,000 by 12,000 to convert the capacity.
Keyword family: 8000 btu to tons; 8000 btu to ton; how many tons is 8000 btu

/9000-btu-to-tons/

Title: 9,000 BTU to Tons = 0.75 [Formula + Chart]
Description: Convert 9,000 BTU/hr to 0.75 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 9,000 BTU to Tons
Answer copy: 9,000 BTU/hr equals 0.75 refrigeration tons. Divide 9,000 by 12,000 to convert the capacity.
Keyword family: 9000 btu to tons; 9000 btu to ton; how many tons is 9000 btu

/10000-btu-to-tons/

Title: 10,000 BTU to Tons = 0.8333 [Formula + Chart]
Description: Convert 10,000 BTU/hr to 0.8333 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 10,000 BTU to Tons
Answer copy: 10,000 BTU/hr equals 0.8333 refrigeration tons (rounded). Divide 10,000 by 12,000 to convert the capacity.
Keyword family: 10000 btu to tons; 10000 btu to ton; how many tons is 10000 btu

/12000-btu-to-tons/

Title: 12,000 BTU to Tons = 1 [Formula + Chart]
Description: Convert 12,000 BTU/hr to 1 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 12,000 BTU to Tons
Answer copy: 12,000 BTU/hr equals 1 refrigeration tons. Divide 12,000 by 12,000 to convert the capacity.
Keyword family: 12000 btu to tons; 12000 btu to ton; how many tons is 12000 btu

Use the shared H2 and internal-link structure on page 19. Rounded titles for 8,000 and 10,000 BTU/hr must be clarified by the on-page answer; preserve full precision in the engine.

19 / READY-TO-IMPLEMENT SEO

Numeric shortcuts 6–10

/18000-btu-to-tons/

Title: 18,000 BTU to Tons = 1.5 [Formula + Chart]
Description: Convert 18,000 BTU/hr to 1.5 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 18,000 BTU to Tons
Answer copy: 18,000 BTU/hr equals 1.5 refrigeration tons. Divide 18,000 by 12,000 to convert the capacity.
Keyword family: 18000 btu to tons; 18000 btu to ton; how many tons is 18000 btu

/24000-btu-to-tons/

Title: 24,000 BTU to Tons = 2 [Formula + Chart]
Description: Convert 24,000 BTU/hr to 2 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 24,000 BTU to Tons
Answer copy: 24,000 BTU/hr equals 2 refrigeration tons. Divide 24,000 by 12,000 to convert the capacity.
Keyword family: 24000 btu to tons; 24000 btu to ton; how many tons is 24000 btu

/30000-btu-to-tons/

Title: 30,000 BTU to Tons = 2.5 [Formula + Chart]
Description: Convert 30,000 BTU/hr to 2.5 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 30,000 BTU to Tons
Answer copy: 30,000 BTU/hr equals 2.5 refrigeration tons. Divide 30,000 by 12,000 to convert the capacity.
Keyword family: 30000 btu to tons; 30000 btu to ton; how many tons is 30000 btu

/36000-btu-to-tons/

Title: 36,000 BTU to Tons = 3 [Formula + Chart]
Description: Convert 36,000 BTU/hr to 3 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 36,000 BTU to Tons
Answer copy: 36,000 BTU/hr equals 3 refrigeration tons. Divide 36,000 by 12,000 to convert the capacity.
Keyword family: 36000 btu to tons; 36000 btu to ton; how many tons is 36000 btu

/42000-btu-to-tons/

Title: 42,000 BTU to Tons = 3.5 [Formula + Chart]
Description: Convert 42,000 BTU/hr to 3.5 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 42,000 BTU to Tons
Answer copy: 42,000 BTU/hr equals 3.5 refrigeration tons. Divide 42,000 by 12,000 to convert the capacity.
Keyword family: 42000 btu to tons; 42000 btu to ton; how many tons is 42000 btu

Use the shared H2 and internal-link structure on page 19. Rounded titles for 8,000 and 10,000 BTU/hr must be clarified by the on-page answer; preserve full precision in the engine.

19 / READY-TO-IMPLEMENT SEO

Numeric shortcuts 11–15

/48000-btu-to-tons/

Title: 48,000 BTU to Tons = 4 [Formula + Chart]
Description: Convert 48,000 BTU/hr to 4 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 48,000 BTU to Tons
Answer copy: 48,000 BTU/hr equals 4 refrigeration tons. Divide 48,000 by 12,000 to convert the capacity.
Keyword family: 48000 btu to tons; 48000 btu to ton; how many tons is 48000 btu

/60000-btu-to-tons/

Title: 60,000 BTU to Tons = 5 [Formula + Chart]
Description: Convert 60,000 BTU/hr to 5 refrigeration tons. See the calculation, cooling kW and nearby capacities, or enter a different rating.
H1: 60,000 BTU to Tons
Answer copy: 60,000 BTU/hr equals 5 refrigeration tons. Divide 60,000 by 12,000 to convert the capacity.
Keyword family: 60000 btu to tons; 60000 btu to ton; how many tons is 60000 btu

/1-ton-to-btu/

Title: 1 Ton to BTU = 12,000 BTU/hr [Quick Chart]
Description: 1 refrigeration tons equals 12,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 1 Ton to BTU Converter
Answer copy: 1 refrigeration tons equals 12,000 BTU per hour. Multiply 1 by 12,000 to calculate the capacity.
Keyword family: 1 ton to btu; 1 tons to btu; 1 ton in btu

/1-5-ton-to-btu/

Title: 1.5 Ton to BTU = 18,000 BTU/hr [Quick Chart]
Description: 1.5 refrigeration tons equals 18,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 1.5 Ton to BTU Converter
Answer copy: 1.5 refrigeration tons equals 18,000 BTU per hour. Multiply 1.5 by 12,000 to calculate the capacity.
Keyword family: 1.5 ton to btu; 1.5 tons to btu; 1.5 ton in btu

/2-ton-to-btu/

Title: 2 Ton to BTU = 24,000 BTU/hr [Quick Chart]
Description: 2 refrigeration tons equals 24,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 2 Ton to BTU Converter
Answer copy: 2 refrigeration tons equals 24,000 BTU per hour. Multiply 2 by 12,000 to calculate the capacity.
Keyword family: 2 ton to btu; 2 tons to btu; 2 ton in btu

Use the shared H2 and internal-link structure on page 19. Rounded titles for 8,000 and 10,000 BTU/hr must be clarified by the on-page answer; preserve full precision in the engine.

19 / READY-TO-IMPLEMENT SEO

Numeric shortcuts 16–20

/2-5-ton-to-btu/

Title: 2.5 Ton to BTU = 30,000 BTU/hr [Quick Chart]
Description: 2.5 refrigeration tons equals 30,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 2.5 Ton to BTU Converter
Answer copy: 2.5 refrigeration tons equals 30,000 BTU per hour. Multiply 2.5 by 12,000 to calculate the capacity.
Keyword family: 2.5 ton to btu; 2.5 tons to btu; 2.5 ton in btu

/3-ton-to-btu/

Title: 3 Ton to BTU = 36,000 BTU/hr [Quick Chart]
Description: 3 refrigeration tons equals 36,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 3 Ton to BTU Converter
Answer copy: 3 refrigeration tons equals 36,000 BTU per hour. Multiply 3 by 12,000 to calculate the capacity.
Keyword family: 3 ton to btu; 3 tons to btu; 3 ton in btu

/3-5-ton-to-btu/

Title: 3.5 Ton to BTU = 42,000 BTU/hr [Quick Chart]
Description: 3.5 refrigeration tons equals 42,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 3.5 Ton to BTU Converter
Answer copy: 3.5 refrigeration tons equals 42,000 BTU per hour. Multiply 3.5 by 12,000 to calculate the capacity.
Keyword family: 3.5 ton to btu; 3.5 tons to btu; 3.5 ton in btu

/4-ton-to-btu/

Title: 4 Ton to BTU = 48,000 BTU/hr [Quick Chart]
Description: 4 refrigeration tons equals 48,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 4 Ton to BTU Converter
Answer copy: 4 refrigeration tons equals 48,000 BTU per hour. Multiply 4 by 12,000 to calculate the capacity.
Keyword family: 4 ton to btu; 4 tons to btu; 4 ton in btu

/5-ton-to-btu/

Title: 5 Ton to BTU = 60,000 BTU/hr [Quick Chart]
Description: 5 refrigeration tons equals 60,000 BTU/hr. Check the formula, compare nearby cooling capacities and edit the value in the free converter.
H1: 5 Ton to BTU Converter
Answer copy: 5 refrigeration tons equals 60,000 BTU per hour. Multiply 5 by 12,000 to calculate the capacity.
Keyword family: 5 ton to btu; 5 tons to btu; 5 ton in btu

Use the shared H2 and internal-link structure on page 19. Rounded titles for 8,000 and 10,000 BTU/hr must be clarified by the on-page answer; preserve full precision in the engine.

20 / READY-TO-IMPLEMENT SEO

Supporting routes 1–4

/tons-to-btu/ / P0

Title: Tons to BTU Calculator [12,000 BTU/hr per Ton]
Description: Convert refrigeration tons to BTU/hr instantly. Check common AC capacities, see the multiply-by-12,000 formula and compare cooling ratings.
H1: Tons to BTU Calculator
Opening copy: 1 ton = 12,000 BTU/hr. Enter a refrigeration-ton rating to see its equivalent capacity.
Target: tons to btu, ton to btu calculator
H2s: Reverse formula | Common ton ratings | Capacity vs sizing

/kw-to-tons/ / P1

Title: kW to Tons Calculator [3.517 kW per Ton]
Description: Convert cooling kilowatts to refrigeration tons with a clear formula and examples. Use cooling capacity, not electrical input power.
H1: kW to Tons Calculator
Opening copy: Enter cooling kW. Divide by approximately 3.516853 to calculate refrigeration tons.
Target: kw to tons, cooling kw to ton
H2s: Cooling kW explained | Conversion example | Electrical input differs

/tons-to-kw/ / P1

Title: Tons to kW Calculator [Cooling Power Explained]
Description: Convert refrigeration tons to cooling kW. See the 3.516853 kW-per-ton factor and learn why cooling output differs from electricity use.
H1: Tons to kW Calculator
Opening copy: One refrigeration ton equals approximately 3.516853 cooling kW.
Target: tons to kw, refrigeration tons to kilowatts
H2s: Tons to kW formula | Common ratings | Estimate input power

/btu-to-kw/ / P1

Title: BTU to kW Calculator [Capacity + Formula]
Description: Convert BTU/hr to cooling kW instantly. View a worked example and common ratings, with a clear distinction between capacity and electricity use.
H1: BTU/hr to kW Calculator
Opening copy: Multiply BTU/hr by 0.0002930710701722222 to obtain cooling kW.
Target: btu to kw, btu hr to kw
H2s: BTU/hr to kW formula | Example conversion | What the result means

Canonical: self. Link to main converter plus the most relevant inverse or next-step route. Do not publish until the described functionality is available.

20 / READY-TO-IMPLEMENT SEO

Supporting routes 5–8

/kw-to-btu/ / P1

Title: kW to BTU Calculator [1 kW = 3,412.14 BTU/hr]
Description: Convert thermal kW to BTU/hr with an editable calculator, formula and examples. Electrical input requires efficiency to infer AC cooling capacity.
H1: kW to BTU/hr Calculator
Opening copy: One thermal kilowatt equals approximately 3,412.142 BTU/hr.
Target: kw to btu, kilowatts to btu hr
H2s: kW to BTU/hr formula | Common values | Thermal vs electrical

/btu-per-minute-to-tons/ / P1

Title: BTU/min to Tons Calculator [200 BTU/min = 1 Ton]
Description: Convert BTU per minute to refrigeration tons. Normalize the time unit, see a worked calculation and compare equivalent BTU/hr and cooling kW.
H1: BTU per Minute to Tons Calculator
Opening copy: Multiply BTU/min by 60, then divide by 12,000. Equivalently, divide BTU/min by 200.
Target: btu min to ton, btu per minute to tons
H2s: Minute-to-hour step | Worked example | Reverse conversion

/btu-per-second-to-tons/ / P1

Title: BTU/s to Tons Calculator [Time-Unit Formula]
Description: Convert BTU per second to refrigeration tons with the correct time factor. See BTU/hr and cooling kW alongside a worked example.
H1: BTU per Second to Tons Calculator
Opening copy: Multiply BTU/s by 0.3 to obtain refrigeration tons.
Target: btu s to tons, btu sec to ton
H2s: Second-to-hour step | Formula | Check your time unit

/btu-to-ton-hours/ / P1

Title: BTU to Ton-Hours [12,000 BTU = 1 Ton-Hour]
Description: Convert thermal energy in BTU to ton-hours. Add a duration to calculate average refrigeration tons and understand energy versus cooling capacity.
H1: BTU to Ton-Hours Calculator
Opening copy: Divide total BTU by 12,000 for ton-hours. Add time only when you need average capacity.
Target: btu to ton hours, ton hour to btu
H2s: Thermal energy formula | Add duration | Energy vs power

Canonical: self. Link to main converter plus the most relevant inverse or next-step route. Do not publish until the described functionality is available.

20 / READY-TO-IMPLEMENT SEO

Supporting routes 9–12

/batch-btu-to-tons/ / P0

Title: Batch BTU to Tons [Convert up to 200 Values]
Description: Paste up to 200 cooling-capacity values, convert them to tons and cooling kW, and download a CSV with row-level validation. No login required.
H1: Batch BTU to Tons Converter
Opening copy: Paste one capacity per line and choose the unit shared by the list.
Target: batch btu to tons; opportunity beyond supplied exact terms
H2s: Paste format | Export columns | Fix invalid rows

/room-ac-size-calculator/ / P1

Title: Room AC Size Calculator [BTU Estimate + Method]
Description: Estimate room-air-conditioner BTU/hr using area, sunlight and occupancy. See the sizing method and understand when a whole-home load calculation is needed.
H1: Room AC Size Calculator
Opening copy: Get a preliminary room-AC capacity estimate using a published sizing table and visible adjustments.
Target: ac btu calculator, btu per square foot, room ac size
H2s: Enter room details | Method and limits | When to use Manual J

/ac-tonnage-by-model-number/ / P0 guide / P2 lookup

Title: AC Tonnage by Model Number [Nameplate Guide]
Description: Find the cooling capacity on your AC nameplate. Learn when model-number codes are reliable and verify supported product families with source documents.
H1: Find AC Tonnage by Model Number
Opening copy: Start with the rated cooling capacity. Use a model code only when its manufacturer documentation is verified.
Target: how to determine ac tonnage from model number, ac tonnage lookup
H2s: Read the nameplate | Supported model families | Unknown model help

/ac-power-cost-calculator/ / P1

Title: AC Power & Cost Calculator [EER or COP Method]
Description: Estimate electrical input and running cost from cooling capacity plus EER or COP, or use known watts. See the assumptions behind every result.
H1: AC Electrical Power and Cost Calculator
Opening copy: Cooling capacity is not electrical input. Add an efficiency rating or enter known input watts to estimate consumption.
Target: ac power consumption; adjacent demand not quantified in CSV
H2s: Efficiency inputs | Electrical energy and cost | Estimate limitations

Canonical: self. Link to main converter plus the most relevant inverse or next-step route. Do not publish until the described functionality is available.

21 / SEARCH IMPLEMENTATION

Main-page SEO and technical contract

Element	Exact implementation
URL / canonical	/btu-to-tons-calculator/ (or homepage on a dedicated domain; choose one).
Title	BTU to Tons Calculator [12,000 BTU/hr = 1 Ton]
Meta description	Convert BTU/hr to tons instantly. Compare AC capacities, use common shortcuts and see cooling kW, clear formulas and a downloadable batch table.
H1 / entity focus	BTU to Tons Calculator. Main entities: BTU/hr, refrigeration ton, cooling capacity, thermal kW. Keep mass and fuel conversions outside this entity set.
Structured data	WebPage + BreadcrumbList; use genuine site Organization data. Optional WebApplication/SoftwareApplication describes the real tool; no invented ratings.
Rich-result limits	Google software-app eligibility requires rating/review data as specified in its documentation. Do not fabricate it to clear a warning. FAQ rich results ended in May 2026. [S8–S9]
Crawling / rendering	Server-render useful initial text, chart and examples. Use normal anchor links. Calculator JS must not be required to discover the main content.
URL variants	Use fragment state for shares; canonical remains clean URL. Do not expose crawlable arbitrary input URLs. Normalize legacy query URLs; do not block a URL whose noindex must be read.
Sitemap / redirects	Only canonical 200-status indexable pages in XML sitemap. Redirect singular aliases and old equivalents with 301. Unknown routes return 404.
Performance budget	Project targets: LCP ≤ 2.5s, INP ≤ 200ms, CLS ≤ 0.1 at 75th percentile. Reserve result space; lazy-load only secondary tools. [S10]

Measurement: events: valid_conversion, preset_selected, compare_completed, batch_exported, sizing_completed, unknown_model. Record unit/mode and error category, not raw pasted values. In Search Console, group queries by route intent and compare CTR while accounting for position and device.

22 / DELIVERY PLAN

Build priorities and further recommendations

Phase	Deliverables	Release gate
P0 / focused launch	Shared engine; desktop/mobile UI; 20 curated numeric candidates; reverse page; batch; comparison; chart; copy; nameplate reading guide.	Calculation tests pass; metadata matches visible features; keyboard/mobile review; indexation rules verified.
P1 / useful extensions	Cooling kW routes; /min and /s; ton-hours; room sizing; power/cost; load-report conversion panel.	Independent HVAC review of method, labels, units and limits; clear separation of estimate versus conversion.
P2 / evidence-backed depth	Manufacturer-family lookup; expanded definitions; optional additional numeric pages based on observed demand.	Documented dataset ownership; exact family scope; regression examples; real search/usage evidence.

Further essentials

1. Show rating basis. Nominal, rated and measured capacity are different. Let users label the source of a rating, without changing the math.
2. Portable AC clarity. Explain that ratings from different test methods may not be directly comparable; do not invent a universal SACC-to-ASHRAE multiplier.
3. Multi-zone caution. A sum of indoor-unit nameplate capacities is not proof of outdoor-unit delivered capacity. If a totalizer is added later, label it a nameplate sum.
4. Technical scope. Defer CFM/airflow, hydronic GPM, cooling-tower tons and heating-loss calculators until their assumptions and source methods are separately specified.

Recommended first growth move

Launch the precise converter and the best-supported shortcuts first. Invest next in the batch/comparison workflow and trustworthy explanations. A large combined page is not automatically better SEO, and additional controls should not slow down a simple conversion.

Editorial owner: verify each keyword-to-page assignment before publishing. **Developer:** maintain one formula engine and one metadata registry. **HVAC reviewer:** validate sizing and electrical guidance. **SEO owner:** monitor indexing and improve weak pages before adding more.

Open limitations: the additional FAQ CSV and the unnamed eighth competitor were not supplied. watttoamp.com could not be retrieved in this session, so this report does not verify or approve that site’s existing breaker logic.

23 / EVIDENCE & LIMITATIONS

Source register 1

[C1] Omni Calculator
https://www.omnicalculator.com/physics/btu-to-tons-converter
Converter, definitions, sharing and reviewer information.

[C2] RapidTables
https://www.rapidtables.com/convert/power/btu-to-ton.html
Converter, formula, inverse link and reference table.

[C3] NumberVibe
https://numbervibe.com/calculators/physics/btu-to-tons-converter-calculator
Combined conversion/sizing, presets, unit inputs, costs and comparison table.

[C4] Calculator Academy
https://calculator.academy/btu-to-tonnage-calculator/
Capacity units, examples and table; three-use gate unconfirmed.

[C5] CtrlCalculator
https://ctrlcalculator.com/physics/btu-to-tons-calculator/
Input, examples, chart and time-unit reminder.

[C6] FreeConvert
https://www.freeconvert.com/unit/btus-per-hour-to-tons-of-refrigeration
Automatic-conversion prompt; dynamic detail limited in retrieval.

[C7] CreativeBooster
https://creativebooster.net/blogs/home/btu-to-tons
Reverse tool, numeric-page links and unit-definition issues.

[S1] NIST SI conversion factors
https://www.nist.gov/pml/special-publication-811/nist-guide-si-appendix-b-conversion-factors/nist-guide-si-appendix-b8
International Table BTU and refrigeration-ton conversion basis.

All listed competitor pages were retrieved on 10 October 2026. Search-service crawl freshness varied. Features were assessed from exposed text and controls; hidden interactions and code were not tested.

23 / EVIDENCE & LIMITATIONS

Source register 2

[S2] ENERGY STAR room air conditioners
https://www.energystar.gov/products/room_air_conditioners
Room area table, scope and adjustment guidance.

[S3] ACCA Manual J
https://www.acca.org/standards/technical-manuals/manual-j
Residential load-calculation scope.

[S4] ACCA Manual S
https://www.acca.org/standards/technical-manuals/manual-s
Equipment-selection scope; use manufacturer performance data.

[S5] Carrier 38TKB product data
https://www.shareddocs.com/hvac/docs/1009/public/0e/38tkb-12pd.pdf
Historical family-specific example; nominal/combination ratings and electrical data. Not a current product recommendation.

[S6] NFPA 70 code-development page
https://www.nfpa.org/codes-and-standards/nfpa-70-standard-development/70
Reference destination only; code text unavailable in this retrieval.

[S7] ENERGY STAR light-commercial criteria
https://www.energystar.gov/products/light_comm_heating_cooling/key_product_criteria
EER definition: cooling BTU/hr divided by electrical watts.

[S8] Google Search documentation updates
https://developers.google.com/search/updates
May 8, 2026 entry: FAQ rich results stopped May 7, 2026.

[S9] Google software-app structured data
https://developers.google.com/search/docs/appearance/structured-data/software-app
Rating/review and offer requirements for eligibility.

[S10] web.dev Web Vitals
https://web.dev/articles/vitals
Core Web Vitals metrics and good-performance thresholds.

User-provided sources: BTU-to-Tons_Landing-Page_SEO-UIUX-Blueprint.pdf and Keywords BTU to Tons.csv. The old PDF is a planning input, not independent proof of competitor features, traffic, standards compliance or schema eligibility.